

Aim:

To design advanced YARA rules using file size constraints, PE header properties, and logical conditions to improve malware detection accuracy and reduce false positives.

Tools Used:

- YARA
- PE analysis tool (e.g., **PEStudio**)
- Malware sample (Portable Executable file)

Procedure:**1. Analyze Malware Sample**

Using a PE analysis tool like PEStudio, extract:

- Entry Point → Starting execution address of the program
- Imported DLLs & APIs → e.g., `kernel32.dll`, `LoadLibraryA`, `VirtualAlloc`
- Number of Sections → Usually 3–6 for normal executables

2. Identify Suspicious Characteristics

- Unusual section names → `.xyz`, `.packed`, `.evil`
- Packed/obfuscated sections → High entropy
- **Suspicious imports:**
 - `VirtualAlloc`
 - `WriteProcessMemory`
 - `CreateRemoteThread`
 - `LoadLibraryA`

3. Advanced YARA Rule

```
import "pe"

rule Advanced_Malware_Detection
{
  meta:
    description = "Detect suspicious PE files using APIs, sections, and size"
    author = "Student"
    date = "2026-05-04"

  strings:
    // Suspicious API strings
    $api1 = "VirtualAlloc" nocase
    $api2 = "WriteProcessMemory" nocase
    $api3 = "CreateRemoteThread" nocase
    $api4 = "LoadLibraryA" nocase

    // Suspicious section names / packer indicators
    $sec1 = ".packed" nocase
    $sec2 = ".upx" nocase
    $sec3 = ".evil" nocase

  condition:
    // Check for PE file (MZ header)
    uint16(0) == 0x5A4D and

    // File size condition (adjusted for your large files)
    filesize < 300MB and

    // PE structure condition
    pe.number_of_sections > 2 and pe.number_of_sections < 10 and

    // Logical detection (balanced)
    (
      1 of ($api*) or
      any of ($sec*)
    )
}
```

```
PS C:\Users\Ram Haygreve> cd C:\Yaralabs
PS C:\Yaralabs> yara advanced_malware.yar samples\
PS C:\Yaralabs> yara advanced_malware.yar samples\
Advanced_Malware_Detection samples\\MSB.exe
PS C:\Yaralabs> dir samples
```

Directory: C:\Yaralabs\samples

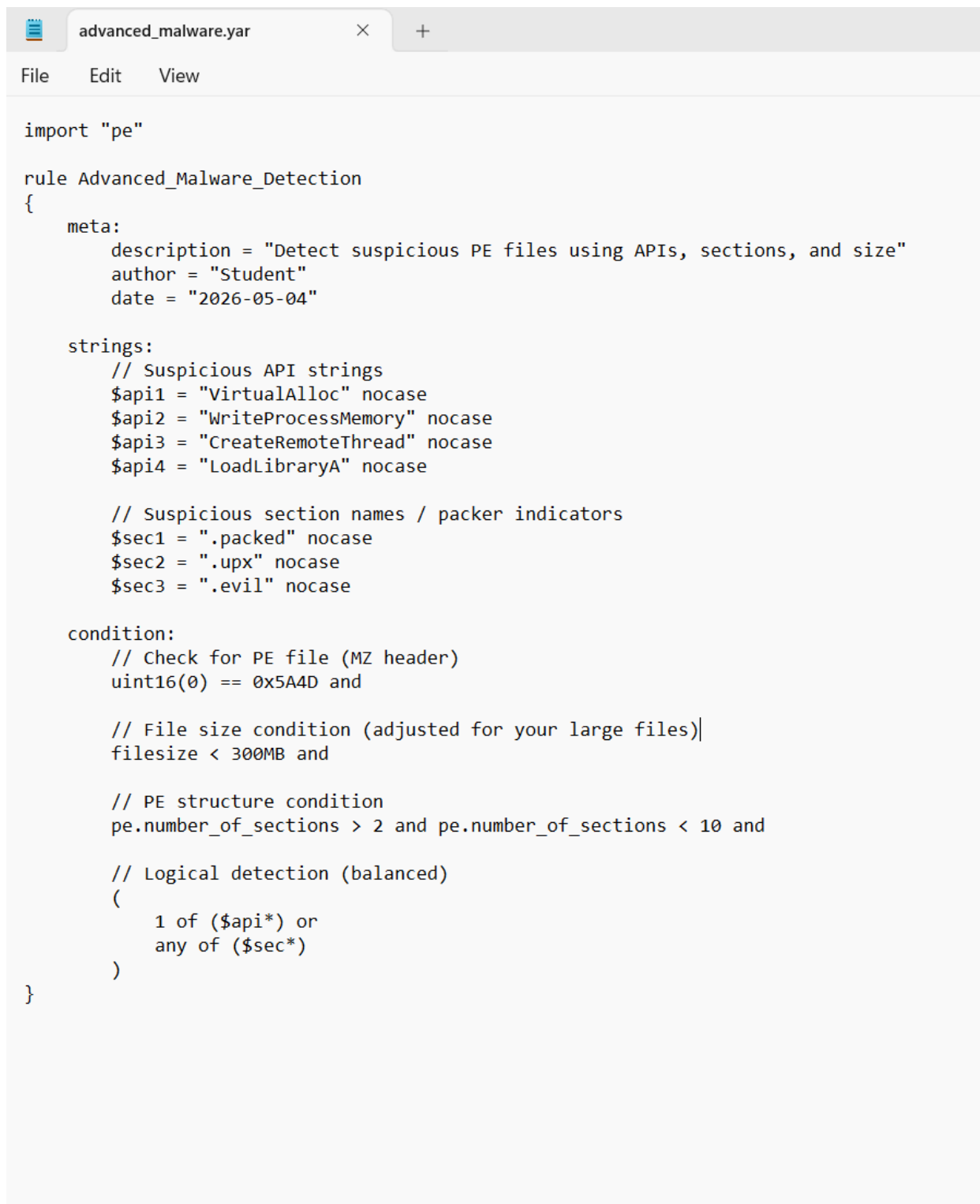
Mode		LastWriteTime	Length	Name
----		-----	-----	----
-a----	01-04-2026	18:32	245644592	Antigravity.exe
-a----	24-04-2026	17:50	242901064	MSB.exe

```
PS C:\Yaralabs> |
```

4. Save the Rule

Save as:

advanced_malware.yar

A screenshot of a YARA rule editor window. The window has a title bar with a file icon, the name 'advanced_malware.yar', and standard window controls (close, maximize, and a plus sign for additional actions). Below the title bar is a menu bar with 'File', 'Edit', and 'View'. The main area contains a YARA rule named 'Advanced_Malware_Detection'. The rule includes a meta section with a description, author, and date. It also has a strings section with suspicious API names and section names. The condition section checks for a valid PE header, file size under 300MB, a minimum of 3 sections, and the presence of suspicious APIs or section names.

```
import "pe"

rule Advanced_Malware_Detection
{
    meta:
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        date = "2026-05-04"

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    condition:
        // Check for PE file (MZ header)
        uint16(0) == 0x5A4D and

        // File size condition (adjusted for your large files)|
        filesize < 300MB and

        // PE structure condition
        pe.number_of_sections > 2 and pe.number_of_sections < 10 and

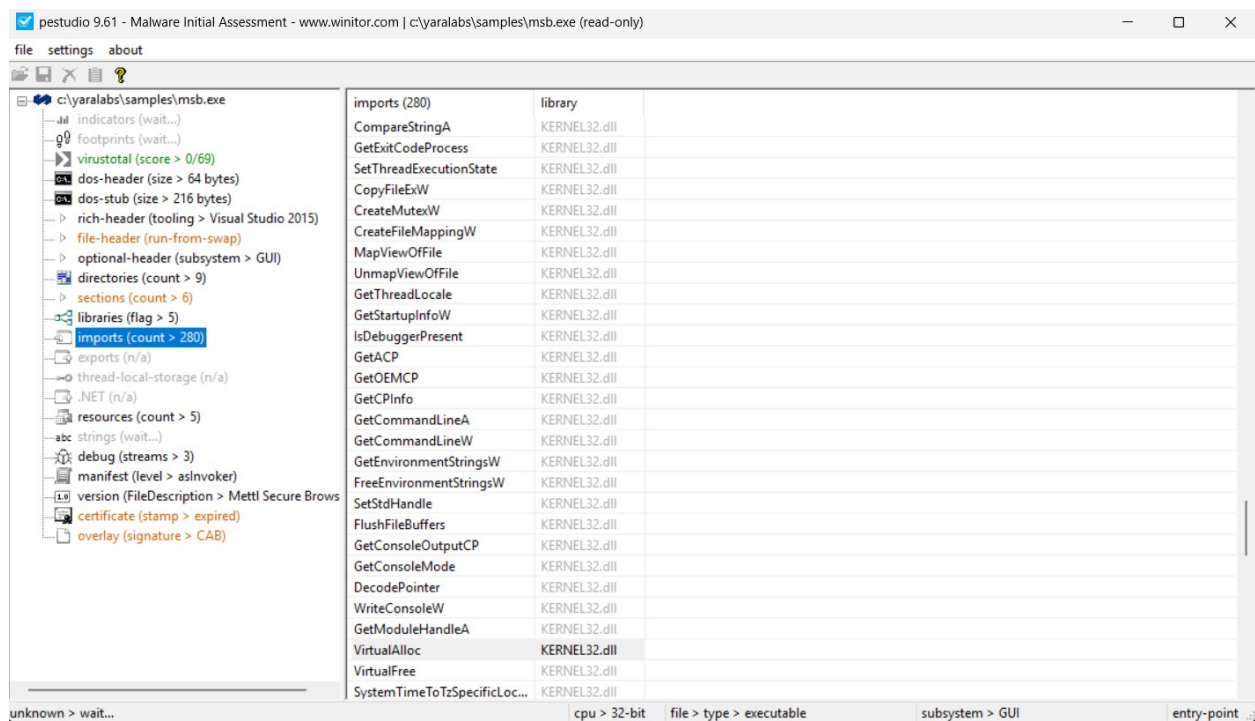
        // Logical detection (balanced)
        (
            1 of ($api*) or
            any of ($sec*)
        )
}
```

5. Scan Files

yara advanced_malware.yar samples/

6. Analyze Results

- Check which files are flagged
- Verify:
 - True positives (actual malware)
 - False positives (benign files detected)



Result:

Hence the experiment was done successfully.